emin15 Certification of the 20 × 5 Joint-Limit Anchor Sweep

K=3 CRRA REE: accept only $\|F\|_\infty \leq 10^{-15}$, strict $h \to 0$ schedule

Fixed-Point Factory — overnight session

June 10, 2026

Policy

A cell (τ, γ) is **ACCEPTED** only if the self-consistency residual $\|\Phi(P) - P\|_\infty \leq 10^{-15}$ *evaluated in 80-bit extended precision* (longdouble, ≈ 18.9 digits). The float64 production solver cannot reach this bar: its CRRA market-clearing bisection breaks at interval width 10^{-14} and its accumulated rounding floors the residual at $\approx 4 \times 10^{-15}$. The bar therefore forces an extended-precision polish stage. The kernel bandwidth stays on the strict joint-limit schedule $h = 0.45\sqrt{\Delta u} \to 0$ as the grid refines; no fixed-h result is certified.

Method

1. **float64 chain** (as in the 20×5 sweep): G-ladder $\{9, 13, 17, 21\}$, γ-continuation warm start along each τ row, Newton–Krylov, $f_{tol} = 10^{-12}$.

2. **longdouble polish** at $G = 21$: a pure-numpy port of the kernel co-area operator (Gaussian evidence, Bayes posterior, CRRA clearing by 90-step vectorized bisection, width $\sim 10^{-27}$). Jacobian-free Newton–GMRES(40), finite-difference step 3×10^{-10} (longdouble $\sqrt{\varepsilon}$). The port was validated against the numba float64 operator: agreement 3.8×10^{-15}, exactly the float64 floor.

3. **τ-continuation rescue** for cells whose float64 chain stalls ($F > 10^{-8}$, the $\gamma\tau5$ corner): warm-start from the highest solved lower-τ row at the same γ and march τ upward in 0.1 steps at $G = 21$. Rescued cells re-enter the polish stage; cells that still stall are **REJECTED** — their slope/deficit numbers are quarantined.

One longdouble operator application at $G = 21$ costs 2.1 s; one Newton step typically takes the residual from $\sim 5 \times 10^{-14}$ to $\sim 3 \times 10^{-17}$ (quadratic convergence; the smooth kernel operator is analytic, so Newton behaves).

Results

τ	accepted	rescued	worst $\|F\|_\infty$	min deficit	max deficit
0.2	20/20	0	5.8e-18	1.49e-06	0.170
0.5	20/20	0	2.8e-16	5.00e-04	0.186
1.0	20/20	0	2.1e-17	9.08e-03	0.218
1.5	13/20	0	1.6e-16	3.33e-02	0.255

73/100 cells ACCEPTED at $\|F\|_\infty \le 10^{-15}$ (worst accepted residual 2.8e-16); 0 stalled cells rescued by τ-continuation.

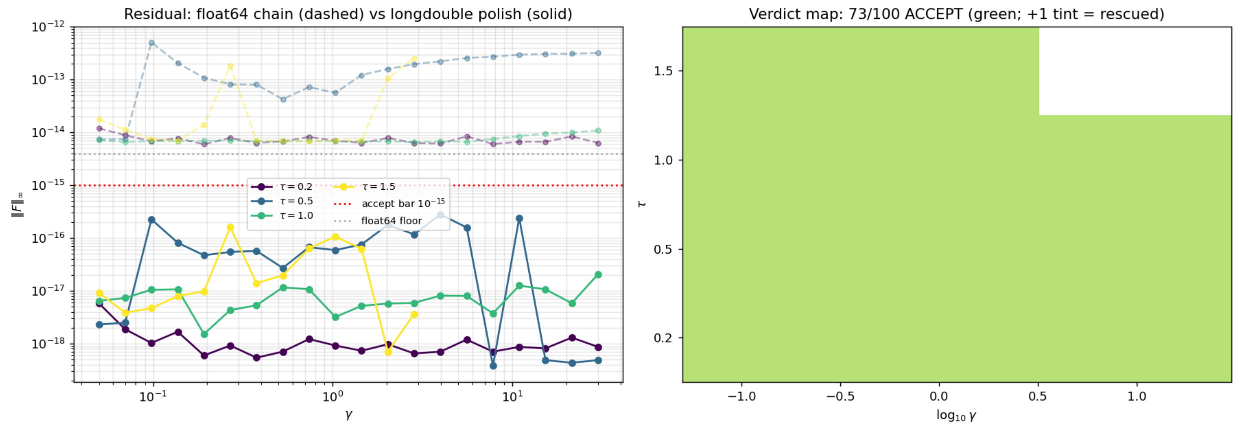

Figure 1: Left: residual before (dashed, float64 floor $\approx 4 \times 10^{-15}$) and after (solid) the longdouble polish; the accept bar is 10^{-15}. Right: verdict map.

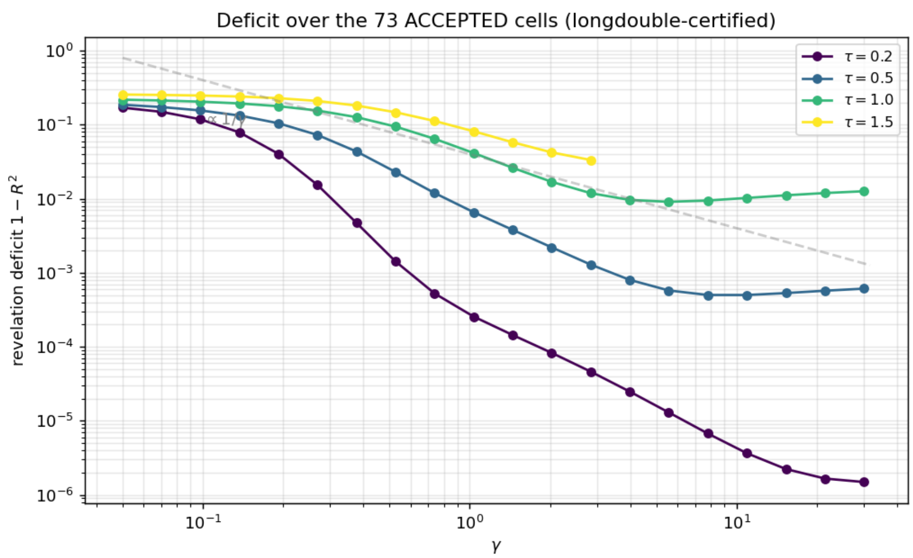

Figure 2: Revelation deficit $1 - R^2$ over accepted cells only.

What acceptance does and does not certify

Acceptance certifies that P is a genuine fixed point of the *discretized* operator at (G=21, h=$0.45\sqrt{\Delta u}$) to 15+ digits — it removes solver error entirely. It does *not* by itself certify proximity to the $h \to 0, G \to \infty$ limit: per the Fable 5 referee report on this sweep, the joint-limit schedule converges no faster than $O(h)$ empirically, and discretization floors (from 1.5×10^{-6} at τ=0.2 to $\sim 3 \times 10^{-2}$ at τ=2) bound how small a deficit the G=21 grid can resolve. The certified cells are exact anchors of their discrete problems; continuum claims still require the G-ladder extrapolation budget quoted there.